

Opponent-conditioned specialization under policy sharing

Train complementary specialists → distill into a shared policy with strategy code z → test whether crossover survives

1 Two regimes

Poles A and B reward different team behaviors. Win rate alone is not enough: the strategy choice must matter.

Regime A

Regime B

2 Dual-branch teachers

Same ATTACK / DEFEND roles on both poles ($k = \text{ceil}(N/3)$). Separate actors π_A and π_B establish complementary play.

π_A

π_B

3 Distill + share

Match teachers on common states into one student. Share encoder or full network; keep discrete strategy input z .

$\pi(a \mid o, z)$

~75% fewer actor params

4 Crossover test

Specialize only if both $\Delta_A > 0$ and $\Delta_B > 0$. Ask whether sharing preserves two-sided specialization.

$\Delta_A = V(A \rightarrow A) - V(B \rightarrow A)$

$\Delta_B = V(B \rightarrow B) - V(A \rightarrow B)$

Finding focus: does opponent-conditioned specialization survive compression into a shared multi-agent policy?